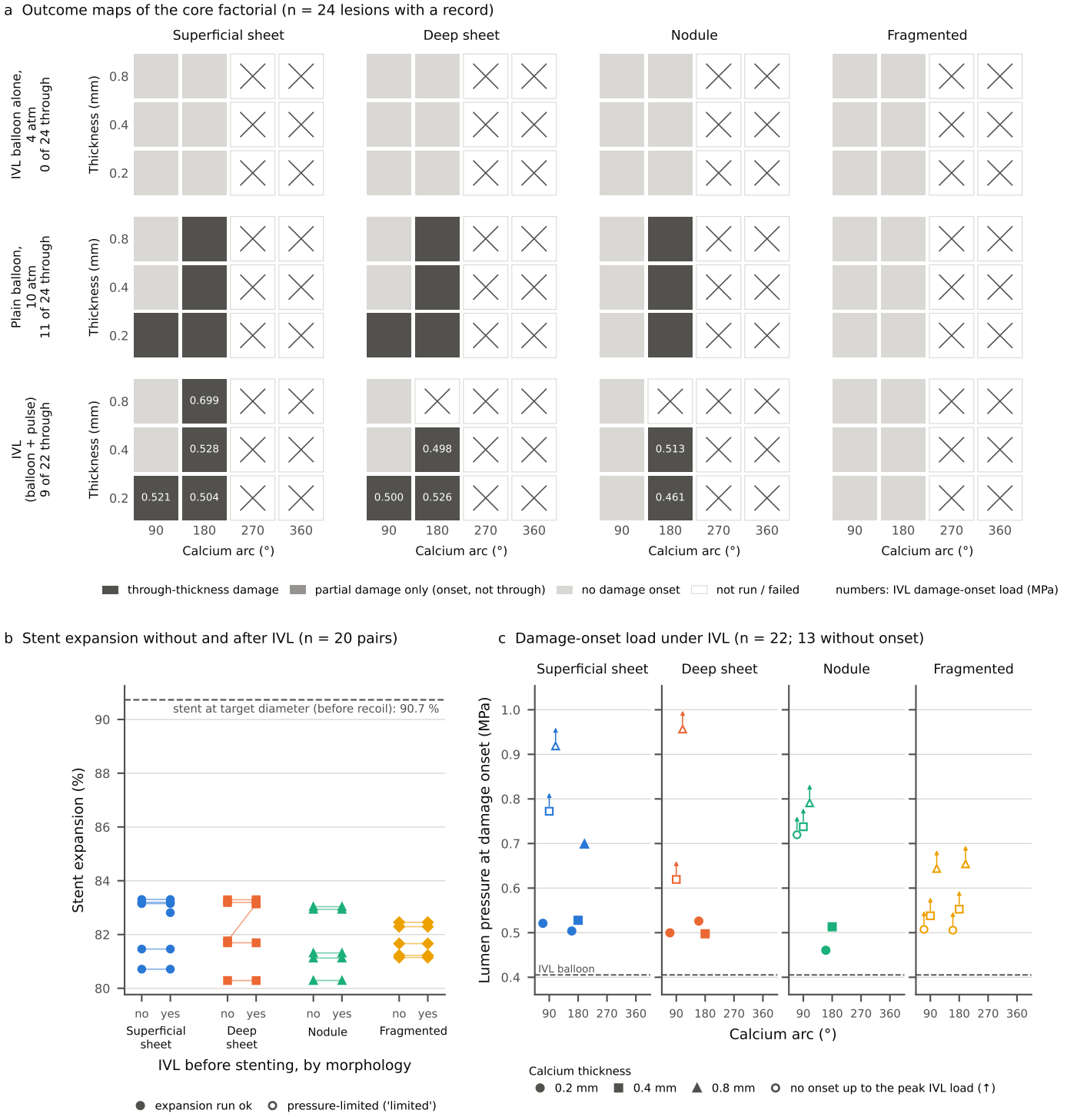
a Outcome maps of the core factorial (n = 24 lesions with a record)
Superficial sheet
Deep sheet
Nodule
Fragmented
IVL balloon alone, 4 atm 0 of 24 through
Plain balloon, 10 atm 11 of 24 through
IVL (balloon + pulse) 9 of 22 through
Thickness (mm)
0.8
0.4
0.2
0.699
0.528
0.521
0.504
0.498
0.500
0.526
0.513
0.461
90
180
270
360
Calcium arc (°)
through-thickness damage
partial damage only (onset, not through)
no damage onset
not run / failed
numbers: IVL damage-onset load (MPa)
b Stent expansion without and after IVL (n = 20 pairs)
stent at target diameter (before recoil): 90.7 %
Stent expansion (%)
80
82
84
86
88
90
no
yes
Superficial sheet
Deep sheet
Nodule
Fragmented
IVL before stenting, by morphology
expansion run ok
pressure-limited ('limited')
c Damage-onset load under IVL (n = 22; 13 without onset)
Lumen pressure at damage onset (MPa)
0.4
0.5
0.6
0.7
0.8
0.9
1.0
IVL balloon
Calcium arc (°)
Calcium thickness
0.2 mm
0.4 mm
0.8 mm
no onset up to the peak IVL load (↑)